

Video Compression - HW1

Custom modifications to the Quantization table -> Bitrate, PSNR

Setup

Baseline `Q` is the standard table from `codec_parameters.py`. Three variants were tested:

- `Q*2` — every entry doubled (coarser quantization)
- `Q/2` — every entry halved (finer quantization)
- `Q_random` — randomized table following the same defining property as `Q` (positive integers, monotonically increasing away from the DC term)

```
Q_random =  
[[ 5,  8, 15, 23, 25, 33, 39, 47],  
 [ 6, 13, 23, 25, 31, 39, 44, 49],  
 [16, 21, 24, 32, 40, 46, 48, 56],  
 [21, 23, 32, 36, 46, 51, 56, 59],  
 [23, 31, 37, 47, 48, 56, 62, 71],  
 [34, 40, 47, 52, 55, 62, 71, 74],  
 [35, 43, 51, 55, 65, 69, 71, 83],  
 [42, 50, 53, 62, 70, 72, 78, 83]]
```

Results

lena_gray.png

Variant	BPP	PSNR (dB)
Q (baseline)	0.8375	35.8084
Q*2	0.5720	33.7049
Q/2	1.2653	37.8650
Q_random	0.8309	35.6145

mandril_gray.png

Variant	BPP	PSNR (dB)
Q (baseline)	1.4545	34.2040
Q*2	1.0449	30.9913
Q/2	1.9433	37.5018
Q_random	1.4153	33.5957

Analysis

Quantization compresses the images by reducing the precision of the DCT coefficients. With `Q*2`, the quantization is coarser. For example, assume $Q(u,v) = 10$, the output range spans from $[LB/10, UB/10]$. With

$Q(u,v) = 20$, the output range spans from $[LB/20, UB/20]$, the information is compressed more, thus lowering the bitrate. With less information, the PSNR is also lower. Same logic applies to $Q/2$, where the quantization is finer, resulting in a higher bitrate and PSNR. As for Q_random , we can observe that the bitrate is similar to the baseline, but the PSNR is slightly lower. This is because Q is the standard quantization table that the industry has refined and settled on for years, so a randomly generated table is not expected to beat it, leading to a slight degradation in quality.

Zigzag vs. Raster-> Bitrate, PSNR

lena_gray.png

Scan order	BPP	PSNR (dB)
Zigzag	0.8375	35.8084
Raster	1.0120	35.8084

mandril_gray.png

Scan order	BPP	PSNR (dB)
Zigzag	1.4545	34.2040
Raster	1.7970	34.2040

Analysis

The scanning order does not affect the PSNR, because after quantization only Huffman coding is applied, which is a lossless compression method. Only bitrate, which shows how efficient the information is compressed, is impacted. In the implementation, the only thing that influences the compression is the trailing zeros. For zigzag scanning, those coefficients divided by large quantization values are at the end of the sequence, and more likely become zero due to the large divisor. The information is mostly stored in the first few coefficients. With raster scanning, the advantage of trailing zeros is less likely to occur, resulting in larger BPP.

Two stage 1D DCT vs. Direct 2D DCT -> Time, bitrate

DCT computation time

Image	Two-stage 1D DCT	Direct 2D DCT	Speedup
lena_gray.png	0.0146 s	0.0225 s	1.55x
mandril_gray.png	0.0141 s	0.0222 s	1.57x

Bitrate (full encode pipeline)

Image	DCT method	BPP
lena_gray.png	Two-stage 1D	0.8375
lena_gray.png	Direct 2D	0.8375

mandril_gray.png	Two-stage 1D	1.4545
mandril_gray.png	Direct 2D	1.4545

Analysis

Two stage 1D DCT and Direct 2D DCT are mathematically equivalent, only floating point errors might cause slight differences in the output, thus the bitrate is the same. Direct 2D DCT has $O(N^4)$ complexity, while two stage 1D DCT has $O(2N^3)$ complexity. This is why two stage 1D DCT is faster than direct 2D DCT.